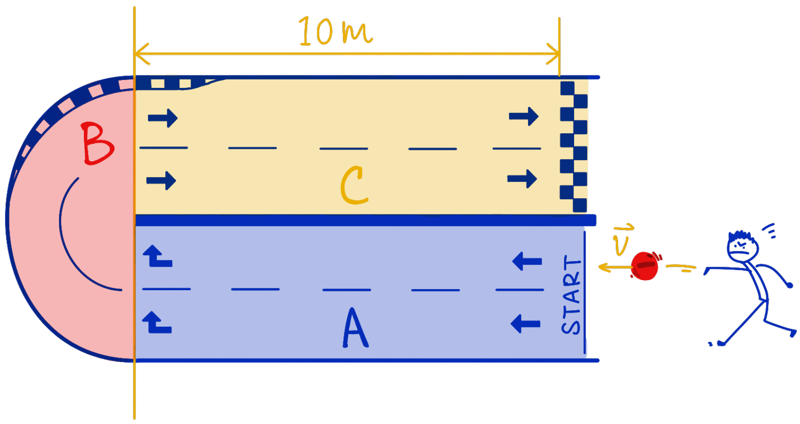
Physics 5C Final Exam Review Session

Department of Physics and Astronomy, UCLA

Problem 1: Jinyan is designing a 2-dimensional racetrack for charged particles. The racetrack consists of three components:

A. Region A is an empty space, where charges are launched into;
B. Region B has a uniform magnetic field, where charges can make a U-turn;
C. Region C has a uniform electric field, where charges sprint towards the finish line.

Neglect gravity and all types of friction in this problem.



Part (a) If the charges launched into region A are all negative, find the direction of the magnetic field in region B such that the charges will turn and go towards region C, then find the direction of the electric field in region C such that it accelerates charges directly towards the finish line.

Part (b) Does the magnetic field in region B change the speed of any of the charges? What type of trajectory do the negative charges follow when they pass through region B?

Part (c) Draw on the diagram the expected trajectory of a negative charge launched into the racetrack. Specify the change in direction and change in speed.

Part (d) A $q_1 = -1\,\text{C}$ charge with mass $m_1 = 2\,\text{kg}$ is launched into region A with initial velocity $1\,\text{m/s}$ to the left. If the magnitude of the magnetic field in region B is $5\,\text{T}$, find the radius of the semi-circular path that the charge will go through in region B.

Part (e) After the charge q_1 has made a U-turn in region B, it enters region C and the electric field will accelerate it. Find the minimal strength of the electric field such that the charge can cross the line with a speed of $10\,\text{m/s}$. Assume that the distance it travels in region C is $10\,\text{m}$. [1]

Part (f) If the strength of the magnetic field in region B and the strength of the electric field in region C are fixed, does the mass of the charge affect its final speed when crossing the line? What about the magnitude of the charge?

[1] There are two different approaches you can take to solve this part - one using kinematics and the other one using conservation of energy. One of them will be much easier in terms of computation.

Problem 2: Now assume that the charges that get launched into the racetrack are electrons being knocked out from a metal. The electrons are knocked out by some photons hitting the metal, and the photons are in turn emitted by some energy-level transition in a Hydrogen atom.

Part (a) Explain in your own words what is going on here. In particular, how do the energy levels of the Hydrogen atom determine the energy of the emitted photon? How would the work function (binding energy) of the metal affect the emission of electrons? How is the speed of the electron determined?

Part (b) If the work function of the metal is $11\,\mathrm{eV}$, find the minimal excited state (minimal energy level n) of the Hydrogen atom such that its transition to the ground state ($n = 1$) can emit a photon that can knock electrons out from the metal.

Part (c) Jinyan is trying to prepare a Hydrogen atom that is in the excited state as calculated in part (b). He is thinking maybe he can do that by using a laser with the "correct" wavelength to excite the atom. Why is this possible? What will be the "correct" wavelength of the laser to excite the Hydrogen atom from its ground state to the correct excited state found in part (b)?

Part (d) Using the laser with the wavelength as found in part (c), and preparing the Hydrogen atoms in their excited states as found in part (b), find the maximal speed of the electrons ejected from the metal.

Part (e) Find the de Broglie wavelength of the electrons from the metal that have their maximal speed.

Part (f) If we want more electrons to be ejected from the metal, what can we do?

Problem 3: A circular wire loop is placed in a uniform constant magnetic field. The direction of the magnetic field is perpendicular to the plane of the circular loop. If the enclosed area of the circular loop is magically changing, find the direction of the induced current in the loop (enumerate all of the possible cases, that is, when $\vec{B}$ is into or out of the page, when area is becoming larger or smaller).

You should ask yourself the following questions before going for the final exam, but sorry, I don't have time to type up their answers ...

---

1. What are conductors, insulators, batteries, resistors, and capacitors?
2. What are source charges and test charges? What is a current?
3. If the electric potential difference across two points is ΔV, how does the energy of a test charge change when it goes across the two points?
4. If a test charge is traveling in an electric field, how to compute the electric force it experiences? How is the force affected by the sign of the charge?
5. How should one calculate the equivalent resistance and equivalent capacitance in a circuit? How is the power of the resistors in the circuit calculated?
6. What is an RC circuit? What are the "limiting cases" in an RC circuit?
7. If I place a test charge at rest in a magnetic field, what is the magnetic force it experiences? What if the test charge is initially moving? If I change the sign of the charge, how does that affect the direction of the magnetic force it experiences?
8. How should one find the direction of a magnetic field produced by straight wires, wire loops, and solenoids? How can one generate current using a magnetic field?
9. What is an electromagnetic wave? How is that related to photons? How can I figure out the direction of propagation of an electromagnetic wave?
10. What is the photoelectric effect? What is the double-slit experiment for electrons? What is the particle-wave duality? What is the Heisenberg uncertainty principle?
11. What is the difference between α, β, and γ decays? How do the atomic number and mass number of elements change in these processes?
12. After a period of three half-lives of a given atom, how much of the original atom do I have left?